COMSM1201 : Programming in C

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About the Course

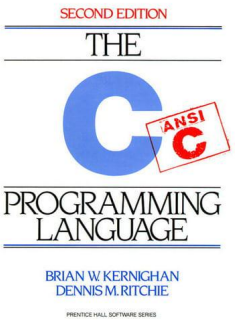
These course notes were originally based on :

C By Dissection (3rd edition)
Al Kelley and Ira Pohl

because I liked arrays being taught late(r).
I've since changed my mind a little & have re-jigged the notes quite heavily over the years.

Resources

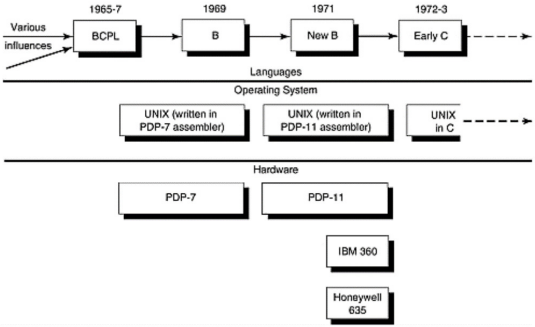
- Free : `https://en.wikibooks.org/wiki/C_Programming`
- A list of more : `https://www.linuxlinks.com/excellent-free-books-learn-c/`
- Whatever you use, make sure it's **C99** that's being taught, not something else e.g. C11 or C++.
- If you fall in love with C and know you're going to use it for the rest of your life, the reference 'bible' is K&R 2nd edition. It's not a textbook for those new to programming, though.



Computer Science Ethos

- Talk to your friends, ask for help, work together.
- Never pass off another persons work as your own.
- Do not pass work to others - either on paper or electronically - even after the submission deadline - it's against our rules.
- If someone takes your code and submits it, we need to investigate where it originated - all students involved will be part of this.
- Don't place your code on publicly accessible sites e.g. github - other students may have extensions etc.

History of C



From **Deep C Secrets** by *Peter Van Der Linden*

- BCPL - Martin Richards
- B - Ken Thomson 1970
- Both of above are *typeless*.
- C - Dennis Ritchie 1972 designed for (& implemented on) a UNIX system.
- K&R C (Kernighan and Ritchie) 1978
- ANSI C
- C99 (COMSM1201)
- C++ Object Oriented Programming (OOP)
- Java (Subset of C++, WWW enabled).

Why C ?

Sep 2024	Sep 2023	Change	Programming Language	Ratings	Change
1	1		Python	20.17%	+6.01%
2	3	↑	C++	10.75%	+0.09%
3	4	↑	Java	9.45%	-0.04%
4	2	↓	C	8.89%	-2.38%
5	5		C#	6.08%	-1.22%
6	6		JavaScript	3.92%	+0.62%
7	7		Visual Basic	2.70%	+0.48%
8	12	⇈	Go	2.35%	+1.16%
9	10	↑	SQL	1.94%	+0.50%
10	11	↑	Fortran	1.78%	+0.49%

https://www.tiobe.com/tiobe-index/

- One of the most commonly used programming languages according to tiobe.com
- Low-level (c.f. Java)
- Doesn't hide nitty-gritty
- Fast ?
- Large parts common to Java

Programming and Software Engineering

- Lectured 4 hours per week (Neill & Kira) for weeks 1-11 (barring week 6)
- Programming (C), data structures, algorithms - searching, sorting, string processing, trees etc.

Assessment

- Weekly (unmarked) exercises that, if completed, should ensure you are able to pass the unit.
- Three assignments (forming a Coursework Portfolio of 70%) and a final, written exam (30%).
- Timings are slightly variable, so don't make any plans in advance - I'll change it if we're going too fast.

Help with Computers

- Any problems with the computers e.g. installing the correct S/W, accessing lab machines : `http://www.bris.ac.uk/it-services/`
- They are also the people to see about passwords etc.
- This page also links to the rather useful Laptop & Mobile Clinic.

Help with the Unit

- All information is available via the Blackboard site (which will point you to other sites including `github.com`, MS Teams etc.)
- Online support is given via the Teams Group.
- The main learning resource is our 6 hour of labs per week staffed by TAs.
- TAs are not allowed to write pieces of code for you, nor undertake detailed bug-fixing of your program.

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Hello World!

```
main( ) {
    extrn a, b, c;
    putchar(a); putchar(b); putchar(c); putchar('!*n');
}

a 'hell';
b 'o, w';
c 'orld';
```

Hello World first seen in: Brian Kernighan, *A Tutorial Introduction to the Language B*, 1972

```c
/* The traditional first program
in honour of Dennis Ritchie
who invented C at Bell Labs
in 1972 */

#include <stdio.h>

int main(void)
{

    printf("Hello, world!\n");
    return 0;

}
```

Execution :

```
Hello, world!
```

Dissecting the 1st Program

- Comments are bracketed by the /* and */ pair.
- #include <stdio.h>
  Lines that begin with a # are called preprocessing directives.
- int main(void)
  Every program has a function called *main()*
- Statements are grouped using braces,

  { ... }
- printf() One of the pre-defined library functions being called (invoked) using a single argument the string :

  "Hello, world!\n"
- The \n means print the single character *newline*.
- Notice all declarations and statements are terminated with a semi-colon.
- return(0) Instruct the Operating System that the function main() has completed successfully.

Area of a Rectangle

```c
#include <stdio.h>

int main(void)
{
   // Compute the area of a rectangle
   int side1, side2, area;
   side1 = 7;
   side2 = 8;
   area = side1 * side2;

   printf("Length of side  1 = %i metres\n", side1);
   printf("Length of side  2 = %i metres\n", side2);
   printf("Area of rectangle = %i metres squared\n", area);
   return 0;
}
```

Execution :

```
Length of side  1 = 7 metres
Length of side  2 = 8 metres
Area of rectangle = 56 metres squared
```

Dissecting the Area Program

- // One line comment.
- #include <stdio.h> Always required when using I/O.
- int side1, side2, area; *Declaration*
- side2 = 8; *Assignment*
- printf() has 2 Arguments. The *control string* contains a %i to indicate an integer is to be printed.

```
1   preprocessing directives
2
3   int main(void)
4   {
5      declarations
6
7      statements
8   }
```

Arithmetic Operators

- + , - , / , *, %
- Addition, Subtraction, Division, Multiplication, Modulus.
- Integer arithmetic discards remainder i.e. 1/2 is 0 , 7/2 is 3.
- Modulus (Remainder) Arithmetic. 7%4 is 3, 12%6 is 0.
- Only available for integer arithmetic.

The Character Type

```c
// Demonstration of character arithmetic
#include <stdio.h>

int main(void)
{
   char   c;

   c = 'A';
   printf("%c ", c);
   printf("%c\n", c+1);
   return 0;
}
```

Execution :

```
A B
```

- The keyword `char` stands for character.
- Used with single quotes i.e. 'A', or '+'.
- Some keyboards have a second single quote the **back quote** ‘
- Note the %c conversion format.

Floating Types

```c
#include <stdio.h>

int main(void)
{

    double x, y;

    x = 1.888888;
    y = 2.111111;

    printf("Sum of x & y is %f.\n", x + y);

    return 0;

}
```

Execution :

```
Sum of x & y is 3.999999.
```

- In C there are three common floating types :
  1. float
  2. double
  3. long double
- The *Working Type* is doubles.

The Preprocessor

- A # in the first column signifies a preprocessor statement.
- #include <file.h> Exchange this line for the entire contents of file.h, which is to be found in a standard place.
- #define PI 3.14159265358979 Replaces all occurrences of PI with 3.14159265358979.
- Include files generally contain other #define's and #include's (amongst other things).

Using printf()

- `printf( fmt-str, arg1, arg2, ...);`

%c	Characters
%i	Integers
%e	Floats/Doubles (Engineering Notation)
%f	Floats/Doubles
%s	Strings

- Fixed-width fields: `printf("F:%7f\n", f);`
  F: 3.0001
- Fixed Precision: `printf("F:%.2f\n", f);`
  F:3.00

Using scanf()

- Similar to `printf()` but deals with *input* rather than *output*.
- `scanf(fmt-str, &arg1, &arg2, ...);`
- Note that the *address* of the argument is required.

`%c`	Characters
`%i`	Integers
`%f`	Floats
`%lf`	Doubles
`%s`	Strings

- Note doubles handled differently than floats.

While Loops

```
while (test is true) {
    statement 1;
        ...
    statement n;
}
```

```c
// Sums are computed.
#include <stdio.h>

int main(void)
{

   int cnt = 0;
   float sum = 0.0, x;
   printf("Input some numbers: ");
   while (scanf("%f", &x) == 1) {
      cnt = cnt + 1;
      sum = sum + x;
   }

   printf("\n%s%5i\n%s%5f\n\n",
          "Count:", cnt, "Sum: ", sum);
   return 0;
}
```

Execution :

```
Input some numbers: 1 5 9 10

Count:     4
Sum: 25.000000
```

Common Mistakes

- Missing "

```
printf("%c\n, ch);
```

- Missing ;

```
a = a + 1
```

- Missing Address in scanf()

```
scanf("%i", a);
```

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Grammar

- C has a grammar/syntax like every other language.
- It has *Keywords*, *Identifiers*, *Constants*, *String Constants*, *Operators* and *Punctuators*.
- Valid Identifiers :
  `k`, `_id`, `iamanidentifier2`, `so_am_i`.
- **Invalid** Identifiers :
  `not#me`, `101_south`, `-plus`.
- Constants :
  `0b10001` (binary), `17` (decimal), `017` (octal), `0x17` (hexadecimal).
- String Constant enclosed in double-quotes :
  `"I am a string"`

Operators

- All operators have rules of both *precedence* and *associativity*.
- $1 + 2 * 3$ is the same as $1 + (2 * 3)$ because $*$ has a higher precedence than $+$.
- The associativity of $+$ is left-to-right, thus $1 + 2 + 3$ is equivalent to $(1 + 2) + 3$.
- Increment and decrement operators : `i++;` is equivalent to `i = i + 1;`
- May also be prefixed `--i;`

```c
#include <stdio.h>

int main(void)
{
    int a, c = 0;
    a = ++c;
    int b = c++;
    printf("%i %i %i\n", a, b, ++c);
    return 0;
}
```

Question : What is the output ?

Assignment

- The = operator has a low precedence and a right-to-left associativity.
- a = b = c = 0; is valid and equivalent to : a = (b = (c = 0));
- i = i + 3; is the same as i += 3;
- Many other operators are possible e.g. -=, *=, /=.

```c
// 1st few powers of 2 are printed.

#include <stdio.h>

int main(void)
{
   int   i = 0, power = 1;

   while (++i <= 10){
      printf("%5i", power *= 2);
   }
   printf("\n");
   return 0;
}
```

Execution :

```
    2    4    8   16   32   64  128  256
512 1024
```

The Standard Library

```c
#include <stdio.h>
#include <stdlib.h>

int main(void)
{
   printf("Randomly distributed integers are printed.\n"
          "How many do you want to see?  ");
   int n;
   if(scanf("%i", &n) == 1){
      for (int i = 0; i < n; ++i) {
         if (i % 4 == 0){
            printf("\n");
         }
         printf("%12i", rand());
      }
      printf("\n");
      return 0;
   }
   return 1;
}
```

Execution :

```
Randomly distributed integers are printed.
How many do you want to see?  11

  1804289383   846930886  1681692777  1714636915
  1957747793   424238335   719885386  1649760492
   596516649  1189641421  1025202362
```

- Definitions required for the proper use of many functions such as `rand()` are found in `stdlib.h`.
- Do not mistake these header files for the libraries themselves !

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Comparisons

<	less than
>	greater than
<=	less than or equal to
>=	greater than or equal to
==	**equal to**
!=	not equal to
!	not
&&	logical AND
||	logical OR

- Any relation is either *true* or *false*.
- Any non-zero value is *true*.
- (a < b) returns the value *0* or *1*.
- (i == 5) is a **test** not an **assignment**.
- (!a) is either *true (1)* or *false (0)*.
- (a && b) is *true* if both a and b are *true*.
- Single & and | are *bitwise* operators not comparisons - more on this later.

Short-Circuit Evaluation

```
if(x >= 0.0 && sqrt(x) < 10.0){

..... /* Do Something */

}
```

It's not possible to take the `sqrt()` of a negative number. Here, the `sqrt()` statement is never reached if the first test is *false*. In a logical AND, once any expression is *false*, the whole must be *false*.

The if() Statement

Strictly, you don't need braces if there is only one statement as part of the if :

```
if (expr)
    statement
```

If more than one statement is required :

```
if (expr) {
    statement-1
        .
        .
        .
    statement-n
}
```

However, we will **always** brace them, even if it's not necessary.

Adding an else statement :

```
if (expr) {
    statement-1
        .
        .
        .
    statement-n
}
else{
    statement-a
        .
        .
        .
    statement-e
}
```

A Practical Example of if:

```c
#include <stdio.h>

int main(void)
{
   int   x, y, z;

   printf("Input three integers:  ");
   if(scanf("%i%i%i", &x, &y, &z) != 3){
      printf("Didn't get 3 numbers?\n");
      return 1;
   }
   int min;
   if (x < y){
      min = x;
   }
   // Nasty, dropped braces:
   else
      min = y;
   if (z < min){
      min = z;
   }
   printf("The minimum value is %i\n", min);
   return 0;
}
```

Execution :

```
Input three integers:  5 7 -4
The minimum value is -4
```

The while() Statement

```
while(expr)
   statement
```

This, as with the for loop, may execute compound statements :

```
while(expr){
   statement-1
        .
        .
        .
   statement-n
}
```

However, we will **always** brace them, even if it's not necessary.

```
1  // Simple while countdown
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7
8     int n = 9;
9
10    while(n > 0){
11       printf("%i ", n);
12       n--;
13    }
14    printf("\n");
15    return 0;
16 }
```

Execution :

```
9 8 7 6 5 4 3 2 1
```

The for() Loop

This is one of the more complex and heavily used means for controlling execution flow.

```
for( init ; test; loop){
   statement-1
        .
        .
        .
   statement-n
}
```

and may be thought of as :

```
init;
while(test){
   statement-1
        .
        .
        .
   statement-n
   loop;
}
```

In the for() loop, note :

- Semi-colons separate the three parts.
- Any (or all) of the three parts could be empty.
- If the test part is empty, it evaluates to *true*.
- for(;;){ a+=1; } is an infinite loop.

A Triply-Nested Loop

```c
// Triples of integers that sum to N
#include <stdio.h>

#define N 7

int main(void)
{
    int    cnt = 0, i, j, k;

    for(i = 0; i <= N; i++){
        for(j = 0; j <= N; j++){
            for(k = 0; k <= N; k++){
                if(i + j + k == N){
                    ++cnt;
                    printf("%3i%3i%3i\n", i, j, k);
                }
            }
        }
    }
    printf("\nCount: %i\n", cnt);
    return 0;
}
```

Output :

```
  0   0   7
  0   1   6
  0   2   5
  0   3   4
  0   4   3
  0   5   2
  0   6   1
  0   7   0
```

etc.

```
  4   3   0
  5   0   2
  5   1   1
  5   2   0
  6   0   1
  6   1   0
  7   0   0
```

Count: 36

The Comma Operator

This has the lowest precedence of all the operators in C and associates left-to-right.

```
a = 0 , b = 1;
```

Hence, the for loop may become quite complex :

```
for(sum = 0, i = 1; i <= n; ++i){
   sum += i;
}
```

An equivalent, but more difficult to read expression :

```
for(sum = 0 , i = 1; i <= n; sum += i , ++i);
```

Notice the loop has an empty body, hence the semicolon.

The do-while() Loop

```
do {
   statement-1
        .
        .
        .
   statement-n
} while ( test );
```

Unlike the while() loop, the do-while() will always be executed at least once.

```
1  // Simple do-while countdown
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7
8     int n = 9;
9
10    /* This program always prints at least one
11       number, even if n initialised to 0 */
12    do{
13       printf("%i ", n);
14       n--;
15    }while(n > 0);
16    printf("\n");
17    return 0;
18 }
```

Execution :

```
9 8 7 6 5 4 3 2 1
```

The switch() Statement

```
switch (val) {
   case 1 :
      a++;
      break;
   case 2 :
   case 3 :
      b++;
      break;
   default :
      c++;
}
```

- The val must be an integer.
- The break statement causes execution to jump out of the loop. No break statement causes execution to 'fall through' to the next line.
- The default label is a catch-all.

The switch() Statement

```c
/* A Prime number can only be divided
   exactly by 1 and itself */

#include <stdio.h>

int main(void)
{

   int i, n;
   do{
      printf("\nEnter a number from 2 - 9 : ");
      n = scanf("%i", &i);
   }while( (n!=1) || (i<2) || (i>9) );
   switch(i){
      case 2:
      case 3:
      case 5:
      case 7:
         printf("That's a prime!\n");
         break;
      default:
         printf("That is not a prime!\n");
   }
   return 0;
}
```

Execution :

```
Enter a number from 2 - 9 : 1

Enter a number from 2 - 9 : 0

Enter a number from 2 - 9 : 10

Enter a number from 2 - 9 : 3
That's a prime!
```

The Conditional (?) Operator

As we have seen, C programers have a range of techniques available to reduce the amount of typing :

```
expr1 ? expr2 : expr3
```

If expr1 is *true* then expr2 is executed, else expr3 is evaluated.

```
1  #include <stdio.h>
2
3  int main(void)
4  {
5     int   x, y, z;
6
7     printf("Input three integers:  ");
8     if(scanf("%i%i%i", &x, &y, &z) != 3){
9        printf("Didn't get 3 numbers?\n");
10       return 1;
11    }
12    int min;
13    min = (x < y) ? x : y;
14    min = (z < min) ? z : min;
15    printf("The minimum value is %i\n", min);
16    return 0;
17 }
```

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Simple Functions

```c
#include <stdio.h>

int min(int a, int b);

int main(void)
{

    int j, k, m;

    printf("Input two integers: ");
    scanf("%i%i", &j, &k);
    m = min(j, k);
    printf("\nOf the two values %i and %i, " \
    "the minimum is %i.\n\n", j, k, m);
    return 0;

}

int min(int a, int b)
{
    if (a < b)
        return a;
    else
        return b;
}
```

Execution :

```
Input two integers: 5 2

Of the two values 5 and 2, the minimum is 2.
```

- Execution begins, as normal, in the `main()` function.
- The function *prototype is shown* at the top of the file. This allows the compiler to check the code more thoroughly.
- The function is defined between two braces.
- The function `min()` returns an `int` and takes two `int`'s as arguments. These are copies of `j` and `k`.
- The `return` statement is used to return a value to the calling statement.

Call-by-Value

In the following example, a function is passed an integer using call by value:

```c
#include <stdio.h>

void fnc1(int x);

int main(void)
{
    int x = 1;

    fnc1(x);
    printf("%i\n", x);
}

void fnc1(int x)
{
    x = x + 1;
}
```

Execution :

```
1
```

- The function does not change the value of `x` in `main()`, since `a` in the function is effectively only a **copy** of the variable.
- A function which has no return value, is declared `void` and, in other languages, might be termed a *procedure*.
- Most parameters used as arguments to functions in C are copied - this is known as *call-by-value*. We'll see the alternative, *call-by-reference*, later.

Testing

```c
#include <stdio.h>

int numfactors(int f);

int main(void)
{

   int n = 12;
   printf("Number of factors in %i is %i\n", \
          n, numfactors(n));
   return 0;
}

int numfactors(int k)
{

   int count = 0;

   for(int i=1; i<=k; i++){
      if( (k%i)==0) {
         count++;
      }
   }
   return count;
}
```

- This is a (not very good) function to compute the number of factors a number has.
- A factor is a number by which a larger (whole/integer) number can be divided.
- 12 has 6 factors: $1, 2, 3, 4, 6$ and 12 itself.
- How do we know the program works though ?
- Running it ?

  `Number of factors in 12 is 6`
- We need something more automated.

Pre- and Post-Conditions

```c
#include <stdio.h>
#include <assert.h>

int numfactors(int f);

int main(void)
{

    int n = 12;
    printf("Number of factors in %i is %i\n", \
           n, numfactors(n));
    return 0;
}

int numfactors(int k)
{

    int count = 0;

    assert(k >= 1); // Avoid trying zero
    for(int i=1; i<=k; i++){
       if( (k%i)==0) {
          count++;
       }
    }
    assert(count <= k);
    return count;
}
```

- Pre-conditions check the inputs to functions, typically their arguments.
- Post-conditions check the returns from functions.
- An assert simple states some test that **ought** to be true. If not, the program aborts with an error.
- There's a sense that this is somehow *safer*, but we haven't exactly done much testing on it to ensure the correct answers are returned.

Assert Testing

```c
#include <stdio.h>
#include <assert.h>

int numfactors(int f);

int main(void)
{
    assert(numfactors(17) == 2);
    assert(numfactors(12) == 6);
    assert(numfactors(6) == 4);
    assert(numfactors(1) == 1);
    assert(numfactors(0) == 0); // ?
    return 0;
}

int numfactors(int k)
{

    int count = 0;

    for(int i=1; i<=k; i++){
       if( (k%i)==0) {
          count++;
       }
    }
    return count;
}
```

- We will use assert testing in this style **every** time we write a function.
- These tests tend to get quite long, so we generally collect them in a function called `test()` which itself is called from `main()`.
- If there is no error, there is no output from this program.
- If you `#define NDEBUG` before the `#include <assert.h>`, all assertions are ignored, allowing them to be used during development and switched off later.

Self-test : Multiply

- Write a simple function `int mul(int a, int b)` which multiples two integers together **without** the use of the multiply symbol in C (i.e. the $*$)
- Use iteration (a loop) to achieve this.
- 7×8 is computed by adding 7, eight times.
- Use `assert()` calls to test it thoroughly - I've given you some to get you started.

```
1   /* Try to write mult(a,b) without using
2      any maths cleverer than addition.     */
3
4   #include <stdio.h>
5   #include <assert.h>
6
7   int mult( int a,  int b);
8   void test(void);
9
10  int main(void)
11  {
12     test();
13
14     return 0;
15  }
16
17  int mult( int a,  int b)
18  {
19
20  // To be completed
21
22  }
23
24  void test(void)
25  {
26     assert(mult(5,3) == 15);
27     assert(mult(3,5) == 15);
28     assert(mult(0,3) == 0);
29     assert(mult(3,0) == 0);
30     assert(mult(1,8) == 8);
31     assert(mult(8,1) == 8);
32  }
```

Program Layout

It is normal for the main() function to come first in a program :

```
#include <stdio.h>
#include <stdlib.h>

list of function prototypes

int main(void)
{
    . . . . .
}

int f1(int a, int b)
{
    . . . . .
}

int f2(int a, int b)
{
    . . . . .
}
```

However, it is theoretically possible to avoid the need for function prototypes by defining a function before it is used :

```
#include <stdio.h>
#include <stdlib.h>

int f1(int a, int b)
{
    . . . . .
}

int f2(int a, int b)
{
    . . . . .
}

int main(void)
{
    . . . . .
}
```

We will **never** use this second approach - put main() first with the prototypes above it.

Replacing Functions with Macros

```c
#include <stdio.h>

#define MIN(A, B) ((A)<(B)?(A):(B))

int main(void)
{

    int j, k, m;

    printf("Input two integers: ");
    scanf("%i%i", &j, &k);
    m = MIN(j, k);
    printf("Minimum is %i\n", m);
    return 0;

}
```

Execution :

```
Input two integers: 5 2
Minimum is 2
```

- There's sometimes a (tiny) time penalty for using functions.
- The contents of the functions are saved onto a special stack, so that when you return to the function, its variables and state can be restored.
- https://en.wikipedia.org/wiki/Call_stack
- Historically, for small functions that needed to be fast, programmers might have #define a macro.
- There's a "double evaluation" problem though - what happens if we used `m = MIN(j++, k++);` ?
- This is expanded to `((j++)<(k++)?(j++):(k++))` which is **not** what was intended.

The inline modifier

- In C99 the inline modifier was introduced https://en.wikipedia.org/wiki/Inline_function

  > *... serves as a compiler directive that suggests (but does not require) that the compiler substitute the body of the function inline by performing inline expansion, i.e. by inserting the function code at the address of each function call, thereby saving the overhead of a function call.*

- Since most compilers have the ability to do this for you, you never need to do this.

```c
#include <stdio.h>

static inline int min(int a, int b);

int main(void)
{

    int j, k, m;

    printf("Input two integers: ");
    scanf("%i%i", &j, &k);
    m = min(j, k);
    printf("Minimum is %i\n", m);
    return 0;

}

inline int min(int a, int b)
{
    if (a < b)
        return a;
    else
        return b;
}
```

Execution :

```
Input two integers: 5 2
Minimum is 2
```

Factorials via Iteration

- A repeated computation computation is normally achieved via *iteration*, e.g. using for():
- Here we compute the factorial of a number - the factorial of 4, written as 4!, is simply $4 \times 3 \times 2 \times 1$.
- Obviously, we'd do more assert tests in the full version.

```c
#include <stdio.h>
#include <assert.h>

int fact(int a);

int main(void)
{
    assert(fact(4) == 24);
    assert(fact(1) == 1);
    assert(fact(0) == 1);
    assert(fact(10) == 3628800);
    return 0;

}

int fact(int a)
{

    int tot = 1;
    for(int i=1; i<=a; i++){
       tot *= i;
    }
    return tot;

}
```

Factorials via Recursion (Advanced)

- We could achieve the same result using recursion.
- The factorial of 4 can be thought of as $4 \times 3!$
- A recursive function calls *itself* - there may be many versions of the same function 'alive' at the same time during execution.

```c
#include <stdio.h>
#include <assert.h>

int fact(int a);

int main(void)
{
    assert(fact(4) == 24);
    assert(fact(1) == 1);
    assert(fact(0) == 1);
    assert(fact(10) == 3628800);
    return 0;

}

int fact(int a)
{

    if(a > 0)
       return ( a * fact(a - 1) );
    else
       return 1;

}
```

Self-test : Multiply (Advanced)

- Write a simple function `int mul(int a, int b)` which multiples two integers together **without** the use of the multiply symbol in C (i.e. the *)
- Use recursion to achieve this.
- Use `assert()` calls to test it thoroughly.

```c
/* Try to write mult(a,b) without using
   any maths cleverer than addition.     */

#include <stdio.h>
#include <assert.h>

int mult( int a,  int b);
void test(void);

int main(void)
{
    test();

    return 0;
}

int mult( int a,  int b)
{

// To be completed

}

void test(void)
{
    assert(mult(5,3) == 15);
    assert(mult(3,5) == 15);
    assert(mult(0,3) == 0);
    assert(mult(3,0) == 0);
    assert(mult(1,8) == 8);
    assert(mult(8,1) == 8);
}
```

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Fundamental Data types

- **[** unsigned | signed **]**
- **[** long | short **]**
- **[** char | int | float | double **]**
- The use of `int` implies signed int without the need to state it.
- Likewise `unsigned short` means `unsigned short int`.

Type	Minimum size (bits)	Format specifier
char	8	%c
signed char	8	%c (or %hhi for numerical output)
unsigned char	8	%c (or %hhu for numerical output)
short short int signed short signed short int	16	%hi or %hi
unsigned short unsigned short int	16	%hu
int signed signed int	16	Identical for printf : %i or %d For scanf, %d input decimal only For scanf, %i auto-detects format (octal,decimal,hex)
unsigned unsigned int	16	%u
long long int signed long signed long int	32	%ld or %li
unsigned long unsigned long int	32	%lu
long long long long int signed long long signed long long int	64	%lli or %lld
unsigned long long unsigned long long int	64	%llu
float		scanf(): %f, %g, %e, %a
double		%lf, %lg, %le, %la
long double		%Lf, %Lg, %Le, %La

Binary Storage of Numbers

In an unsigned char :

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	1	0	0	1	1	0	0

The above represents :
$1 * 64 + 1 * 8 + 1 * 4 = 76$.

- Floating operations need not be exact.

```
1  #include <stdio.h>
2
3  int main(void)
4  {
5
6      float d = 0.1;
7      printf("%.12f\n", 3.0*d);
8      return 0;
9  }
```

Execution :

```
0.300000004470
```

- Not all floats are representable so are only approximated.
- Since floats may not be stored exactly, it doesn't make sense to try and compare them:

```
if ( d == 0.3 )
```

- Therefore, we don't allow this by explicitly using the compiler warning flag: -Wfloat-equal

sizeof()

To find the exact size in bytes of a type on a particular machine, use sizeof(). On a Dell Windows 10 laptop running WSL:

```c
#include <stdio.h>

int main(void)
{

    printf("char       :%3li\n", sizeof(char));
    printf("short      :%3li\n", sizeof(short));
    printf("long       :%3li\n", sizeof(long));
    printf("unsigned   :%3li\n", sizeof(unsigned));
    printf("long long  :%3li\n", sizeof(long long));
    printf("float      :%3li\n", sizeof(float));
    printf("dbl        :%3li\n", sizeof(double));
    printf("long dbl   :%3li\n", sizeof(long double));
    printf("\n");

    return 0;
}
```

Execution :

```
char       :  1
short      :  2
long       :  8
unsigned   :  4
long long  :  8
float      :  4
dbl        :  8
long dbl   : 16
```

Mathematical Functions

- There are no mathematical functions built into the C language.
- However, there are many functions in the maths library which may linked in using the **-lm** option with the compiler e.g. `gcc math1.c -o math1 -lm`
- Functions include :

  `sqrt()` `pow()` `round()`
  `fabs()` `exp()` `log()`
  `sin()` `cos()` `tan()`
- Most take `doubles` as arguments and return `doubles`.

Casting

```c
/* Compute the Volume of a Sphere
   to the nearest integer       */
#include <stdio.h>
#include <math.h>

#define PI 3.14159265358979323846

int main(void)
{
   double r;
   printf("Enter a radius : ");
   scanf("%lf", &r);
   // Make sure radius is positive
   r = fabs(r);
   double v = 4.0 / 3.0 * PI * pow(r, (double) 3);
   printf("Volume of your ball = %f\n", v);
   printf("Volume of your ball = %.2f\n", v);
   printf("Volume of your ball = %i\n", (int)v);
   printf("Volume of your ball = %.0f\n", v);
   printf("Volume of your ball = %f\n", round(v));
   return 0;
}
```

Execution :

```
Enter a radius : 7.75
Volume of your ball = 1949.816390
Volume of your ball = 1949.82
Volume of your ball = 1949
Volume of your ball = 1950
Volume of your ball = 1950.000000
```

- An explicit type conversion is called a *cast*.
- *If it moves - cast it.* Don't trust the compiler to do it for you !

Storage of Characters

- Characters are stored in the machine as one byte (generally 8-bits storing one of **256** possible values).
- These may be thought of a characters, or very small integers.
- Only a subset of these 256 values are required for the printable characters, space, newline etc.
- Declaration:

```
char c;

c = 'A';
```

or :

```
char c1 = 'A', c2 = '*', c3 = ';' ;
```

- The particular integer used to represent a character is dependent on the encoding used. The most common of these, used on most UNIX and PC platforms, is ASCII.

lowercase	'a'	'b'	'c'	...	'z'
ASCII value	97	98	99	...	122
uppercase	'A'	'B'	'C'	...	'Z'
ASCII value	65	66	67	...	90
digit	'0'	'1'	'2'	...	'9'
ASCII value	48	49	50	...	57
other	'&'	'*'	'+'	...	
ASCII value	38	42	43	...	

Using Characters

- When using printf() and scanf() the formats %c and %i do very different things :

```
char c = 'a'
printf("%c\n", c); /* prints : a  */
printf("%i\n", c); /* prints : 97 */
```

- Hard-to-print characters have an escape sequence i.e. to print a newline, the 2 character escape '\n' is used.

Escape sequence	Hex value	Character
\a	07	Alert (Beep, Bell)
\b	08	Backspace
\e	1B	Escape character
\f	0C	Formfeed Page Break
\n	0A	Newline (Line Feed)
\r	0D	Carriage Return
\t	09	Horizontal Tab
\v	0B	Vertical Tab
\\	5C	Backslash
\'	27	Apostrophe
\"	22	Double quote
\?	3F	Question mark

Using getchar() and putchar()

```c
// Outputs characters twice

#include <stdio.h>

int main(void)
{

    char c;

    do{
       c = getchar();
       putchar(c);
       putchar(c);
    }while(c != '!');
    putchar('\n');

    return 0;
}
```

Execution :

```
abc123!
aabbcc112233!!
```

This has the unfortunate problem of requiring a 'special' character to terminate. More aggressively, the user could terminate by pressing CTRL-C.

```c
// Outputs characters twice

#include <stdio.h>

int main(void)
{

    char c; // char or int ?

    while ((c = getchar()) != EOF) {
       putchar(c);
       putchar(c);
    }
    putchar('\n');

    return 0;
}
```

Execution :

```
abc123
aabbcc112233
```

The end-of-file constant is defined in stdio.h. Although system dependent, -1 is often used. On the UNIX system this is generated when the end of a file being piped is reached, or when CTRL-D is pressed.

Capitalization

```c
// Outputs characters twice

#include <stdio.h>

#define CAPS ('A' - 'a')

int main(void)
{
   int   c;
   while ((c = getchar()) != '!'){
      if (c >= 'a' && c <= 'z'){
          putchar(c + CAPS);
      }
      else{
         putchar(c);
      }
   }
   putchar('\n');

   return 0;
}
```

Execution :

```
Hello World!
HELLO WORLD
```

This is more easily achieved by using some of the definitions found in ctype.h.

Macro	true returned if:
isalnum(int c)	Letter or digit
isalpha(int c)	Letter
iscntrl(int c)	Control character
isdigit(int c)	Digit
isgraph(int c)	Printable (not space)
islower(int c)	Lowercase
isprint(int c)	Printable
ispunct(int c)	Punctuation
isspace(int c)	White Space
isupper(int c)	Uppercase
isxdigit(int c)	Hexadecimal
isascii(int c)	ASCII code

ctype.h

Some useful functions are :

Function/Macro	Returns:
int tolower(int c)	Lowercase c
int toupper(int c)	Uppercase c
int toascii(int c)	ASCII code for c

```c
#include <stdio.h>
#include <ctype.h>

int main(void)
{

   int c;
   while ((c = getchar()) != EOF){
      if (islower(c)){
         putchar(toupper(c));
      }
      else{
         putchar(c);
      }
   }
   putchar('\n');
   return 0;
}
```

```c
#include <stdio.h>
#include <ctype.h>

int main(void)
{

   int c;
   while ((c = getchar()) != EOF){
      /* toupper() returns non-lowercae
         chars unaltered */
      putchar(toupper(c));
   }
   putchar('\n');
   return 0;
}
```

Execution :

```
Hello World!
HELLO WORLD!
```

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Enumerated Types

```
enum day { sun, mon, tue, wed, thu, fri, sat};
```

- This creates a user-defined **type** `enum day`.
- The enumerators are constants of type `int`.
- By default the first (`sun`) has the value 0, the second has the value 1 and so on.

- An example of their use:

```
enum day d1;
 . . .
d1 = fri;
```

- The default numbering may be changed as well:

```
enum fruit{apple=7, pear, orange=3, lemon};
```

- Use enumerated types as constants to aid readability - they are self-documenting.
- Declare them in a header (`.h`) file.
- Note that the type is `enum day`; the keyword `enum` is not enough.

Typedefs

- Sometimes it is useful to associate a particular name with a certain type, e.g.:

  `typedef int colour;`
- Now the type `colour` is synonymous with the type `int`.
- Makes code self-documenting.
- Helps to control complexity when programmers are building complicated or lengthy user-defined types (See Structures later).

Combining typedefs and enums

- Often typedef's are used in conjunction with enumerated types:

```c
#include <stdio.h>
#include <assert.h>

enum day {mon,tue,wed,thu,fri,sat, sun};
typedef enum day day;

day find_next_day(day d);

int main(void)
{
    assert(find_next_day(mon)==tue);
    assert(find_next_day(sat)==sun);
    assert(find_next_day(sun)==mon);
    return 0;
}
```

```c
day find_next_day(day d)
{
    day next_day;
    switch(d){
        case sun:
            next_day = mon;
            break;
        case mon:
            next_day = tue;
            break;
        case tue:
            next_day = wed;
            break;
        case wed:
            next_day = thu;
            break;
        case thu:
            next_day = fri;
            break;
        case fri:
            next_day = sat;
            break;
        case sat:
            next_day = sun;
            break;
        default:
            printf("I wasn't expecting that !\n");
    }
    return next_day;
}
```

Style

```
enum veg {beet, carrot, pea};
typedef enum veg veg;
veg v1, v2;
v1 = carrot;
```

- We can combine the two operations into one:

  ```
  typedef enum veg {beet,carrot,pea} veg;
  veg v1, v2;
  v1 = carrot;
  ```

- Assigning:

  ```
  v1 = 10;
  ```

  is very poor programming style !

Booleans

- Before C99 you might have been tempted to define your own Boolean type:

```c
#include <stdio.h>
#include <assert.h>

typedef enum bool {false, true} bool;

int main(void)
{

    bool b = true;
    if (b){
        printf("It's true!\n");
    }
    else{
        printf("It's false!\n");
    }
    return 0;
}
```

Execution :

```
It's true!
```

- However, we can just use #include <stdbool.h>

```c
#include <stdio.h>
#include <stdbool.h>
#include <assert.h>

int main(void)
{

    bool b = true;
    if (b){
        printf("It's true!\n");
    }
    else{
        printf("It's false!\n");
    }
    return 0;
}
```

Execution :

```
It's true!
```

Fever

Rewrite/complete this code using typedefs and enums to create self-documenting code in any manner you wish.

```c
#include <stdio.h>
#include <assert.h>

// Argument 1 is temperature
// Argument 2 is scale (0=>Celsius, 1=>Fahrenheit)
int fvr(double t, int s);

int main(void)
{
    assert(fvr(37.5, 0)==1);
    assert(fvr(36.5, 0)==0);
    assert(fvr(96.5, 1)==0);
    assert(fvr(99.5, 1)==1);
    return 0;
}

int fvr(double t, int s)
{
}
```

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1D Arrays

- One-Dimensional arrays are declared by a type followed by an identifier with a bracketed constant expression:

  `float x[10];`
  `int k[ARRAY_SIZE];`

- The following, (a variable-length array), is not valid:

  `float y[i*2];`

  we prevent these using the `-Wvla` compiler flag.

- Arrays are stored in contiguous memory, e.g.:

  `int a[5];`



- Arrays are indexed **0** to **n-1**.

```
1  #include <stdio.h>
2
3  #define N 500
4
5  int main(void)
6  {
7
8     /* allocate space a[0]...a[N-1] */
9     int a[N];
10    int i, sum = 0;
11    /* fill array */
12    for (i = 0; i < N; ++i){
13       a[i] = 7 + i * i;
14    }
15    /* print array */
16    for (i = 0; i < N; ++i){
17       printf("a[%i]=%i ", i, a[i]);
18    }
19    /* sum elements */
20    for (i = 0; i < N; ++i){
21       sum += a[i];
22    }
23    /* print sum */
24    printf("\nsum=%i\n", sum);
25    return 0;
26 }
```

1D Arrays : Initialisation

By default, arrays are uninitialised. When they are declared, they may be assigned a value:

```
float x[7] = {-1.1,0.2,2.0,4.4,6.5,0.0,7.7};
```

or,

```
float x[7] = {-1.1, 0.2};
```

the elements 2 ... 6 are set to zero.
Also:

```
int a[] = {3, 8, 9, 1};
```

is valid, the compiler assumes the array size to be 4.

- `a[5] = a[4] + 1;`
- `k[9]++;`
- `n[12+i] = 0;`
- Accessing an array out of bounds will not be identified by the compiler. It may cause an error at run-time. One frequent result is that an entirely unrelated variable is altered.

1D Arrays : Call by Reference

```c
#include <stdio.h>
#include <math.h>
#include <assert.h>
#define MAX 5

// Pass array, AND number of elements
void set_array(int a[MAX], unsigned int len, int n);

int main(void)
{
   int x[MAX] = {2, 3, 3, 3, 3};
   set_array(x, 5, 3); assert(x[0] == 3);
   x[0] = 5; x[1] = 5; x[2] = 5; x[3] = 5; x[4] = 5;
   set_array(x, 5, 4); assert(x[2] == 4);
   set_array(x, 1, 0); assert(x[0] == 0);
   x[0] = 1; x[1] = 2; x[2] = 3;
   set_array(x, 3, 2);
   assert(x[2] == 2); assert(x[3] == 4);
}

// Set all values of array (size len) to n
void set_array(int a[MAX], unsigned int len, int n)
{
   if(len == 0){
      return;
   }
   for(unsigned int i=0; i<len; i++){
     a[i] = n;
   }
}
```

- Here, the array is passed by *Reference* - no copy of the array is made - the function processes the array that was created inside `main()`, despite it apparently having a 'different' name.
- All arrays are passed like this in C - we'll see later when we look at *pointers* why this is the case.

Structures

- A structure type allows the programmer to aggregate components into a single, named variable. Other languages call these *Records* or *Tuples*.
- Each component has individually named members.
- struct employee {
  long id;
  double salary;
  short age;
  };
- struct is a keyword, employee is the structure tag name, and id, salary and age are members of the structure.
- A statement of the form :
  struct employee e1, e2;
  actually creates storage for the variables.
- A member is accessed using the member operator "."
- e1.salary = 35000.2;
  e2.age = 29;
- The member name must be unique within the same structure.
- Arrays of structures are possible, i.e.:
  struct employee team[400];

Arrays of Structures

```c
#include <stdio.h>
#include <stdbool.h>
#include <stdlib.h>
#include <assert.h>

#define SUITS 4
#define PERSUIT 13
#define DECK (SUITS*PERSUIT)
#define SHUFFLE 3

typedef enum {hearts, diamonds, spades, clubs} suit;

struct card {
   suit st;
   int pips;
};
typedef struct card card;

void shuffle_deck(card d[DECK]);
void init_deck(card d[DECK]);
void print_deck(card d[DECK], int n);
void test(void);

int main(void)
{
   card d[DECK];

   test();
   init_deck(d);
   print_deck(d, 7);
   shuffle_deck(d);
   print_deck(d, 7);
   return 0;
}
```

```c
void init_deck(card d[DECK])
{
   for(int i=0; i<DECK; i++){
      // Number 1 .. 13
      d[i].pips = (i%PERSUIT) + 1;
      switch (i/PERSUIT) {
         case hearts: d[i].st = hearts; break;
         case diamonds: d[i].st = diamonds; break;
         case spades: d[i].st = spades; break;
         case clubs: d[i].st = clubs; break;
         // Force an abort ?
         default : assert(false);
      }
   }
}

void shuffle_deck(card d[DECK])
{
  for(int i=0; i<SHUFFLE*DECK; i++){
     int n1 = rand()%DECK;
     int n2 = rand()%DECK;
     card c = d[n1]; d[n1] = d[n2]; d[n2] = c;
  }
}
```

Arrays of Structures

```
void print_deck(card d[DECK], int n)
{
   for(int i=0; i<n; i++){
      switch(d[i].pips){
         case 11:
            printf("Jack");
            break;
         case 12:
            printf("Queen");
            break;
         case 13:
            printf("King");
            break;
         default:
            printf("%2i", d[i].pips);
      }
      switch(d[i].st){
         case hearts :
            printf(" of Hearts\n");
            break;
         case diamonds :
            printf(" of Diamonds\n");
            break;
         case spades:
            printf(" of Spades\n");
            break;
         default :
            printf(" of Clubs\n");
      }
    }
    printf("\n");
}
```

Execution :

```
 1 of Hearts
 2 of Hearts
 3 of Hearts
 4 of Hearts
 5 of Hearts
 6 of Hearts
 7 of Hearts

 4 of Spades
Jack of Spades
 7 of Clubs
 9 of Spades
10 of Spades
 7 of Hearts
 2 of Spades
```

- The `print_deck()` function is clearly messy ! We can simplify this a little when we understand strings.

Testing

```
void test(void)
{
   int n = 0;
   card d[DECK];
   init_deck(d);
   // Direct assignment
   card c = {hearts, 10};
   // 1st element initialised correctly
   assert(d[9].pips == c.pips);
   assert(d[9].st == c.st);
   for(int i=0; i<1000; i++){
      shuffle_deck(d);
      // Happens 1 time in 52 ?
      if((d[0].st == c.st) && (d[0].pips == c.pips)){
         n++;
      }
   }
   // Is this a reasonable test ?
   assert((n > 10) && (n < 30));
}
```

- Note the direct ability to copy a structure.
- You can't compare them using == though.
- Tricky to think of a good test for shuffle_deck.
- You could also typedef away the array, e.g.:

  typedef card deck[DECK];

  void shuffle_deck(deck d);

  But this hides the fact it's an array (which seems odd?)

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Strings

- Strings are 1D arrays of characters.
- Any character in a string may be accessed as an array element.
- The important difference between strings and ordinary arrays is the **end-of-string sentinel** '\0' or null character.
- The string "abc" has a *length* of 3, but its *size* is 4.
- Note 'a' and "a" are different. The first is a character constant, the second is a string with 2 elements 'a' and '\0'.

Initialising Strings :

- `char w[6] = "Hello";`
- `char w[250];`
  `w[0] = 'a';`
  `w[1] = 'b';`
  `w[2] = 'c';`
  `w[3] = '\0';`
- `scanf("%s", w);`
  Removes leading spaces, reads a string (terminated by a space or EOF). Adds a null character to the end of the string.
- `char w[250] = {'a', 'b', 'c', '\0'};`

Unused Letters and string.h

```c
#include <stdio.h>
#include <stdbool.h>
#include <ctype.h>

#define ALPHASIZE 26

int main(void)
{
    char s[100] = "The Quick Brown Fox Leaps" \
                  "Over the Lazy Dog";
    bool used[ALPHASIZE] = {false};
    int i = 0;
    while(s[i]){
        char c = tolower(s[i]);
        if(islower(c)){
            used[c - 'a'] = true;
        }
        i++;
    }
    for(i=0; i<ALPHASIZE; i++){
        if(!used[i]){
            printf("%c has not been used.\n", i+'a');
        }
    }
    return 0;
}
```

Execution :

```
j has not been used.
m has not been used.
```

In #include <string.h> :

```
char *strcat(char dest[], const char src[]);
int strcmp(const char s1[], const char s2[]);
```

- strcat() appends a copy of string src, including the terminating null character, to the end of string dst.
- strcmp() compares two strings byte-by-byte, according to the ordering of your machine's character set. The function returns an integer greater than, equal to, or less than 0, if the string pointed to by s1 is greater than, equal to, or less than the string pointed to by s2 respectively.

More string.h

In #include <string.h> :

```
char *strcpy(char dst[], const char src[]);
unsigned strlen(const char s[]);
```

- strcpy() copies string src to dst including the terminating null character, stopping after the null character has been copied.
- strlen() returns the number of bytes in s, not including the terminating null character.

One way to write the function strlen() :

```
1  #include <stdio.h>
2  #include <assert.h>
3
4  unsigned nstrlen(const char s[]);
5
6  int main(void)
7  {
8      assert(nstrlen("Neill")==5);
9      assert(nstrlen("")==0);
10     assert(nstrlen("\n")==1);
11     assert(nstrlen("abcdef")==nstrlen("fedcba"));
12     return 0;
13 }
14
15 unsigned nstrlen(const char s[])
16 {
17     register unsigned n = 0;
18
19     while(s[n] != '\0'){
20         ++n;
21     }
22     return n;
23 }
```

The snprintf() Function

In #include <string.h> : This is very similar to the function printf(), except that the output is stored in a string rather than written to the output. It is defined as:

int snprintf(string, str-size, control-arg, other args);

For example:

```
int i = 7;
float f = 17.041;
char str[100];
snprintf(str, 100, "%i %f", i, f);
printf("%s\n", str);
```

Outputs : 7 17.041000

This is useful if you need to create a string for passing to another function for further processing.

```
#define SMALLSTR 20
void print_card(char s[BIGSTR], card c)
{

   char pipstr[SMALLSTR];
   char suitstr[SMALLSTR];
   switch(c.pips){
      case 11:
         strcpy(pipstr, "Jack");
         break;
      case 12:
         strcpy(pipstr, "Queen");
         break;
      case 13:
         strcpy(pipstr, "King");
         break;
      default:
         snprintf(pipstr, SMALLSTR, "%2i", c.pips);
   }
   switch(c.st){
      case hearts :
         strcpy(suitstr, "Hearts");
         break;
      case diamonds :
         strcpy(suitstr, "Diamonds");
         break;
      case spades:
         strcpy(suitstr, "Spades");
         break;
      default :
         strcpy(suitstr, "Clubs");
   }
   snprintf(s, BIGSTR, "%s of %s", pipstr, suitstr);
}
```

snprintf() and sscanf()

```c
#define FIRSTCARD " 1 of Hearts"
void test(void)
{
   int n = 0;
   char str[BIGSTR];
   card d[DECK];
   init_deck(d);
   // Direct assignment
   print_card(str, d[0]);
   // 1st element initialised correctly
   assert(strcmp(str, FIRSTCARD)==0);
   for(int i=0; i<1000; i++){
      shuffle_deck(d);
      print_card(str, d[0]);
      // Happens 1 time in 52 ?
      if(strcmp(str, FIRSTCARD)==0){
         n++;
      }
   }
   // Is this a reasonable test ?
   assert((n > 10) && (n < 30));
}
```

```c
// Simple demo of sscanf (and fgets in passing)
#include <stdio.h>
#include <assert.h>

#define BIGSTR 1000
#define SMLSTR 100
#define DAYSINYEAR 365.2425

#include <stdio.h>

int main(void)
{


   printf("Please type your first name and your age\n");
   char bigstr[BIGSTR];
   fgets(bigstr, BIGSTR, stdin);
   char name[SMLSTR];
   int age;
   // Note no "&" before name : passed by reference already
   assert(sscanf(bigstr, "%s %i\n", name, &age)==2);
   printf("%s, you've lived approximately %.0f days\n",
          name, ((double)(age)+0.5)*DAYSINYEAR);
   return 0;

}
```

Execution :

```
Please type your first name and your age
Joe 25
Joe, you've lived approximately 9314 days
```

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Initializing 2D Arrays

A 2D array is declared as follows:

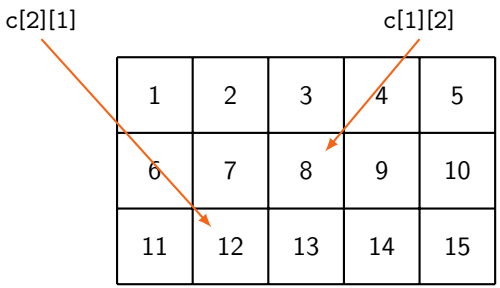
```
#define ROWS 3
#define COLS 5
int a[ROWS][COLS];
```

2D array initialisation :

```
int b[2][3] = {1, 2, 3, 4, 5, 6};
int b[2][3] = {{1, 2, 3}, {4, 5, 6}};
int b[ ][3] = {{1, 2, 3}, {4, 5, 6}};
```

Although 2D arrays are stored in a contiguous block of memory, we may think of them as a 2D rectangle of data.

```
int c[3][5] = {{1,2,3,4,5}, {6,7,8,9,10},
               {11,12,13,14,15}};
```



2D Distance

```c
#include <stdio.h>
#include <math.h>

#define   M   7
#define   N   9

void fillarray(int a[M][N]);

int main(void)
{
   int   a[M][N];

   fillarray(a);
   // Print Array
   for (int j = 0; j < M; j++){
      for(int i = 0; i < N; i++){
         printf("%i", a[j][i]);
      }
      printf("\n");
   }
   printf("\n");
   return 0;
}

void fillarray(int a[M][N])
{
   for (int j = 0; j < M; ++j){
      double y = ((double)j - ((double)(M-1)/2.0));
      for(int i = 0; i < N; ++i){ // Column-first
         double x = ((double)i - ((double)(N-1)/2.0));
         a[j][i] = round(sqrt(x*x + y*y));
      }
   }
}
```

Execution :

```
544333445
443222344
432111234
432101234
432111234
443222344
544333445
```

Cards (again!)

```c
#define SMALLSTR 20
void print_card(char s[BIGSTR], card c)
{
   // Note the +1 below : zero pips not used, but makes easier coding ?
   char pipnames[PERSUIT+1][SMALLSTR] = {"Zero", "One", "Two", "Three",
                                         "Four", "Five", "Six", "Seven",
                                         "Eight", "Nine", "Ten", "Jack",
                                         "Queen", "King"};
   char suitnames[SUITS][SMALLSTR] = {"Hearts", "Diamonds", "Spades", "Clubs"};
   snprintf(s, BIGSTR, "%s of %s", pipnames[c.pips], suitnames[c.st]);
}
```

- The $2D$ arrays of characters here have one string per row.
- They are of a fixed-width, sometime called *ragged-right* or *jagged-right* arrays.

Storage Classes

- **auto**

  auto int a, b, c;
  auto float f;

  Because this is the default, it is seldom used.
- **extern**
  Tells the compiler to look for the variable elsewhere, possibly another file.
- **register**
  Informs the compiler to place the variable in a high-speed memory register if possible, i.e. if there are enough such registers available & the hardware supports this.

```c
#include <stdio.h>
#include <stdlib.h>

void printstuff(void);

#define MAXLOOP 20

int main(void)
{
   int r = rand() % MAXLOOP;
   for(int i=0; i<r; i++){
      printstuff();
   }
   return 0;
}

void printstuff(void)
{
   static int cnt = 0;
   printf("You've been here %i times\n", ++cnt);
}
```

Execution :

```
You've been here 1 times
You've been here 2 times
You've been here 3 times
```

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Call-by-Value

```c
#include <stdio.h>

void changex(int x);

int main(void)
{
   int x = 1;

   changex(x);
   printf("%i\n", x);
   return 0;
}

void changex(int x)
{
   x = x + 1;
}
```

Execution :

```
1
```

- In the program, the function cannot change the value of `v` as defined in `main()` since a **copy** is made of it.
- To allow a function to modify the value of a variable passed to it we need a mechanism known as **call-by-reference**, which uses the **address** of variables (pointers).
- (strictly speaking, we're only simulating call-by-reference by passing an address using call-by-value, but we'll ignore this for now).

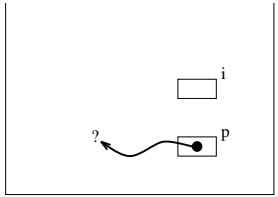
Call-by-Reference

- We have already seen addresses used with `scanf()`. The function call:

  `scanf("%i", &v);`

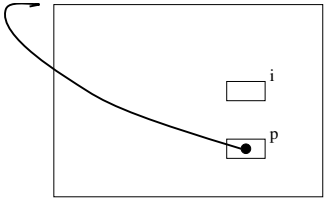
  causes the appropriate value to be stored at a particular address in memory.
- If `v` is a variable, then `&v` is its address, or location, in memory.

- `int i, *p;`
- Here `i` is an `int` and `p` is of type *pointer to int*.
- Pointers have a legal range which includes the special address 0 and a set of positive integers which are the machine addresses of a particular system.

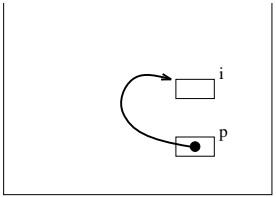


The *NULL* Pointer

- `p = NULL;`

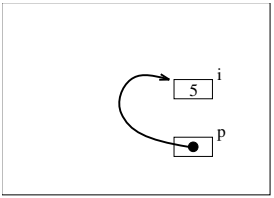


- `p = &i;`



Equivalence of *i* and **p*

- i = 5;




```c
#include <stdio.h>

int main(void)
{

    int i = 5;
    int* p = &i;
    printf("%i\n", *p);
    i = 17;
    printf("%i\n", *p);
    *p = 99;
    printf("%i\n", i);

    return 0;

}
```

Execution :

```
5
17
99
```

scanf Again

```c
#include <stdio.h>

int main(void)
{

    int i;
    int* p = &i;
    printf("Please Type a number : ");
    scanf("%i", &i);
    printf("%i\n", i);
    printf("Please Type a number : ");
    scanf("%i", p);
    printf("%i\n", i);

    return 0;

}
```

Execution :

```
Please Type a number : 70
70
Please Type a number : 3
3
```

- In many ways the dereference operator $*$ is the inverse of the address operator $\&$.

```c
float x = 5, y = 8, *p;
p = &x;
y = *p;
```

- What is this equivalent to ?

The *swap* Function

```c
#include <stdio.h>

void swap(int* p, int* q);

int main(void)
{
    int   a = 3, b = 7;

    // 3 7 printed
    printf("%i %i\n", a, b);
    swap(&a, &b);
    // 7 3 printed
    printf("%i %i\n", a, b);
    return 0;
}

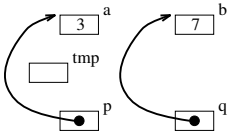
void swap(int* p, int* q)
{
    int   tmp;

    tmp = *p;
    *p = *q;
    *q = tmp;
}
```

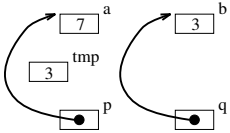
Execution :

```
3 7
7 3
```

- At beginning of function:



- At end of function:



- Remember that the variables *a* and *b* are not in the scope of *swap()*.

Arrays are Pointers ?

- An array name by itself is simply an address (**Array Decay**).
- For instance:

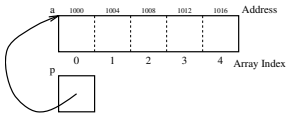
  `int a[5];`
  `int *p;`

  declares an array of 5 elements, and `a` is the address of the start of the array.
- Assigning:

  `p = a;`

  is completely valid and the same as:

  `p = &a[0];`



- To assign `p` to point to the next element, we could either :

  `p = a + 1;`
  `p = &a[1];`
- Notice that `p = a + 1` advances the pointer **4** bytes and not 1 byte. This is because an integer is 4 bytes long and `p` is a pointer to an int.
- we can use the pointer `p` is exactly the same way as normal, i.e.:

  `*p = 5;`

Summing an Array

```c
#include <stdio.h>

#define NUM 5

int sum(int a[]);

int main(void)
{

    int n[NUM] = {10, 12, 6, 7, 2};

    printf("%i\n", sum(n));
    return 0;
}

int sum(int a[])
{
    int tot = 0;

    for(int i=0; i<NUM; i++){
       tot += a[i];
    }
    return tot;
}
```

Execution :

37

```c
#include <stdio.h>

#define NUM 5

int sum(int a[]);

int main(void)
{

    int n[NUM] = {10, 12, 6, 7, 2};

    printf("%i\n", sum(n));
    return 0;
}

int sum(int a[])
{
    int tot = 0;

    for(int i=0; i<NUM; i++){
       tot += *(a + i);
    }
    return tot;
}
```

Execution :

37

```c
#include <stdio.h>

#define NUM 5

int sum(int* p );

int main(void)
{

    int n[NUM] = {10, 12, 6, 7, 2};

    printf("%i\n", sum(n));
    return 0;
}

int sum(int* p )
{
    int tot= 0;

    for(int i=0; i<NUM; i++){
       tot += *p;
       p++;
    }
    return tot;
}
```

Execution :

37

Pointers to Structures

- By default, structures are passed by value (copied) when used as a parameter to a function.
- But, like any other type, we could pass a pointer instead.
- The complication is that to access the elements of a structure via a pointer, we use the "->" operator, and not the ".".

```
void print_deck(card d[DECK], int n)
{
   char str[BIGSTR];
   for(int i=0; i<n; i++){
      print_card(str, &d[i]);
      printf("%s\n", str);
   }
   printf("\n");
}

#define SMALLSTR 20
void print_card(char s[BIGSTR], const card* p)
{
   // Note the +1 below : zero pips not used, but makes easier coding ?
   char pipnames[PERSUIT+1][SMALLSTR] = {"Zero", "One", "Two", "Three",
                                         "Four", "Five", "Six", "Seven",
                                         "Eight", "Nine", "Ten", "Jack",
                                         "Queen", "King"};
   char suitnames[SUITS][SMALLSTR] = {"Hearts", "Diamonds", "Spades", "Clubs"};
   snprintf(s, BIGSTR, "%s of %s", pipnames[p->pips], suitnames[p->st]);
}
```

Nested Structures

```c
#include <stdio.h>

struct dateofbirth {
   unsigned char day;
   unsigned short month;
   unsigned short year;
};
typedef struct dateofbirth dob;

typedef struct {
   char* name;
   dob date;
} person;

void print_byval(person b);
void print_byref(const person* p);

int main(void)
{
   person a = {"Gary", {17, 5, 1999}};
   print_byval(a);
   print_byref(&a);
}

void print_byval(person b)
{
   printf("%s %hhu/%hi/%hi\n", b.name, b.date.day, b.date.month, b.date.year);
}

void print_byref(const person* p)
{
   printf("%s %hhu/%hi/%hi\n", p->name, p->date.day, p->date.month, p->date.year);
}
```

Execution :

```
Gary 17/5/1999
Gary 17/5/1999
```

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String Constants

```c
// A FAILED attempt to
// convert all 'n' chars to 'N'
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <assert.h>

void nify(char s[]);

int main(void)
{

    nify("neill");
    return 0;

}

// In-Place : Swaps all 'n' -> 'N'
void nify(char s[])
{
    for(int i=0; s[i]; i++){
        if(s[i] == 'n'){
            s[i] = 'N';
        }
    }
}
```

- This looks (at first) like a sensible attempt to accept a string and change it *in-place* to capitalise all 'n' characters. It crashes though via a segmentation fault.
- With the usual compile flags we get no more information.
- But using:

  gcc nify1.c -g3 -fsanitize=undefined -fsanitize=address -o nify1

  we find that

  ```
  s[i] = 'N';
  ```

  is the culprit.
- It turns out that in `main()` we have passed a **constant** string to the function. This is in a part of memory that we have read-only permission.

Local Variables

```c
// A FAILED attempt to
// convert all 'n' chars to 'N'
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <assert.h>

#define LINE 500

char* nify(char s[]);

int main(void)
{

    char* s1 = nify("inconveniencing");
    char* s2 = nify("neill");
    assert(strcmp(s2, "Neill")==0);
    assert(strcmp(s1, "iNcoNveNieNciNg")==0);
    return 0;

}

// Local copy : Swaps all 'n' -> 'N'
char* nify(char s[])
{
    char t[LINE];
    strcpy(t, s);
    for(int i=0; t[i]; i++){
       if(t[i] == 'n'){
          t[i] = 'N';
       }
    }
    return t;
}
```

- Now we try to create a copy of the string, and return a pointer to it.
- With the usual compile flags we're told:

  nify2.c: In function 'nify':
  nify2.c:33:11: warning: function returns address of local variable [-Wreturn-local-addr]
  33 | return t;
- The string t is local to nify().
- What happens in this memory when outside the scope of this function is completely undefined.

Static Variables

```c
// A FAILED attempt to
// convert all 'n' chars to 'N'
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <assert.h>

#define LINE 500

char* nify(char s[]);

int main(void)
{

    char* s1 = nify("inconveniencing");
    char* s2 = nify("neill");
    assert(strcmp(s2, "Neill")==0);
    assert(strcmp(s1, "iNcoNveNieNciNg")==0);
    return 0;

}

// Local copy : Swaps all 'n' -> 'N'
char* nify(char s[])
{
    static char t[LINE];
    strcpy(t, s);
    for(int i=0; t[i]; i++){
       if(t[i] == 'n'){
          t[i] = 'N';
       }
    }
    return t;
}
```

- We could just make the local string a static and return it's address couldn't we?
- This only works if we're very careful with the order in which we use the strings.
- This code fails because, in main(), by the time we strcmp(s1, "iNcoNveNieNciNg") the contents of s1 have been overwritten by "Neill".
- The pointers s1 and s2 are the same.

Using *malloc()*

- We must use malloc() instead.
- void* malloc(int n);
  allocates *n* bytes and returns a pointer to the allocated memory. The memory is not initialized.
- Now, when our function is called, a dedicated chunk of memory is allocated from the heap.
- This memory is always in scope until free() is used on it.
- We must free the memory somewhere though, otherwise memory leaks occur.
- We will see calloc() (and perhaps realloc()) later.

```
1  #include <stdlib.h>
2  #include <string.h>
3  #include <assert.h>
4
5  char* nify(char s[]);
6
7  int main(void)
8  {
9     char* s1 = nify("inconveniencing");
10    char* s2 = nify("neill");
11    assert(strcmp(s2, "Neill")==0);
12    assert(strcmp(s1, "iNcoNveNieNciNg")==0);
13    free(s1);
14    free(s2);
15    return 0;
16 }
17
18 // malloc : Swaps all 'n' -> 'N'
19 char* nify(char s[])
20 {
21    int l = strlen(s);
22    char* t  = (char*)malloc(l+1);
23    if(t==NULL){
24       exit( EXIT_FAILURE );
25    }
26    strcpy(t, s);
27    for(int i=0; t[i]; i++){
28       if(t[i] == 'n'){
29          t[i] = 'N';
30       }
31    }
32    return t;
33 }
```

Variable Length Arrays

```c
// This code is not allowed by the -Wvla flag
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <assert.h>

#define WORD 500

int main(void)
{

    printf("Please type a string :\n");
    char s[WORD];
    assert(scanf("%s", s)==1);
    int n = strlen(s) + 1;
    char t[n];
    // Deep copy: character by character
    strcpy(t, s);
    printf("%s %s\n", s, t);
    return 0;

}
```

- Here we duplicate a string into t.
- This is known as a variable length array.
- However, we will always use the -Wvla with the compiler to prevent them.
- There are a number of reasons for this:
  - Some C++ compilers don't accept it.
  - The memory comes off the stack not the heap, and you have no idea if the allocation has worked (it'll just crash if not)
  - https://nullprogram.com/blog/2019/10/27/
- None of these is a problem if we use malloc().

Memory Leaks

```c
// This leaks - but it's not obvious
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <assert.h>

#define WORD 500

int main(void)
{

    printf("Please type a string :\n");
    char s[WORD];
    assert(scanf("%s", s)==1);
    int n = strlen(s);
    /* malloc() returns a pointer to memory that
       you have access to. Note forcing cast. */
    char* t = (char*) malloc(n+1);
    // If no space, returns NULL
    assert(t != NULL);
    // Deep copy: character by character
    strcpy(t, s);
    printf("%s %s\n", s, t);
    return 0;

}
```

- This code appears to work correctly.
- However, it actually **leaks**. The memory allocated was never free()'d.
- This is best found by compiling the program using the sanitizer:

```
==201503==ERROR: LeakSanitizer: detected memory leaks
Direct leak of 6 byte(s) in 1 object(s) allocated from:
    #0 0x737ae50b4887 in ___interceptor__malloc ../../../../src/libsanitizer/
asan/asan__malloc__linux.cpp:145
    #1 0x5845d6ac347c in main (/mnt/c/Users/csnwc/OneDrive - University
of Bristol/Dropbox/Lecturing/COMSM1201/strdup+0x147c)
    #2 0x737ae4429d8f in ___libc__start__call__main ../sysdeps/nptl/libc__start
__call__main.h:58
SUMMARY: AddressSanitizer: 6 byte(s) leaked in 1 allocation(s).
```

free()

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <assert.h>

#define WORD 500

int main(void)
{

    char s[WORD] = "String";
    int n = strlen(s);
    /* malloc() returns a pointer to memory that
       you have access to. Note forcing cast. */
    char* t = (char*) malloc(n+1);
    /* If no space, returns NULL */
    assert(t != NULL);
    /* Deep copy: character by character */
    strcpy(t, s);
    printf("%s %s\n", s, t);
    /* All malloc'd memory must be freed
       to prevent memory leaks */
    free(t);
    return 0;

}
```

- This code is now correct.
- The sanitized code runs without any leak warnings.

Structures with Self-Referential Pointers

```c
// Store a list of numbers
#include <stdio.h>
#include <stdlib.h>
#include <assert.h>

struct data {
    int num;
    struct data* next;
};
typedef struct data data;

int main(void)
{
//        a       b       c

//    |11 -->|17 -->| 5 . |
    data c = {5 , NULL};
    data b = {17, &c};
    data a = {11, &b};

    // print first number
    printf("%i\n", a.num);
    data* p = &a;
    // Can also get to it via p
    printf("%i\n", p->num);
    // Pointer chasing : The Key concept
    p = p->next;
    // We're accessing b, without using it's name
    printf("%i\n", p->num);
    p = p->next;
    // And c
    printf("%i\n", p->num);
    return 0;
}
```

- The structure contains a pointer to something of it's own type (even before we've fully defined the struture itself).
- Here, if p points to a, then p->next->next points to c.

Linked Lists

```c
// Store a list of numbers (length unknown)
#include <stdio.h>
#include <stdlib.h>
#include <assert.h>

#define MAXNUM 20
#define ENDNUM 10

struct data {
   int num;
   struct data* next;
};
typedef struct data data;

void addtolist(data* tail);
void printlist(data* st);

int main(void)
{
   data *p, *start;
   start = p = calloc(1, sizeof(data));
   assert(p);
   p->num = rand()%MAXNUM;
   // Add other numbers to the list
   do{
      addtolist(p);
      p = p->next;
   }while(p->num != ENDNUM);
   printlist(start);
   // Need to free up list - not shown here ...
   return 0;
}
```

```c
// Create some new space and store number in it
void addtolist(data* tail)
{
   tail->next = calloc(1, sizeof(data));
   assert(tail->next);
   tail = tail->next;
   tail->num = rand()%MAXNUM;
}

void printlist(data* st)
{
   while(st != NULL){
      printf("%i ", st->num);
      st = st->next;
   };
   printf("\n");
}
```

Execution :

```
3 6 17 15 13 15 6 12 9 1 2 7 10
```

- calloc() is similar to malloc(), but *clears* the memory is reserves for you. It's passed the number of array cells you wish to create, and the size of each of them.

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File Properties

- They have a name.
- Until a file is opened nothing can be done with it.
- After use a file must be closed.
- Files may be read, written or appended.
- Conceptually a file may be thought of as a stream of characters.

```c
/* Creates a file called helloworld.txt
in the current filespace and writes to it */
#include <stdio.h>
#include <stdlib.h>

int main(void)
{
   // Clobber the file if it exists already
   FILE* fp = fopen("helloworld.txt", "w");
   if(fp == NULL){
      fprintf(stderr, "Cannot open file?\n");
      exit(EXIT_FAILURE);
   }
   fprintf(fp, "Hello World!\n");
   // fscanf() is also available to read in
   fclose(fp);
   return EXIT_SUCCESS;
}
```

Reading and Writing

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#define BIGSTR 500
#define FNAME "helloworld.txt"

int main(void)
{
    FILE* fp = fopen(FNAME, "w");
    if(fp == NULL){
        fprintf(stderr, "Cannot open file %s ?\n", FNAME);
        exit(EXIT_FAILURE);
    }
    fprintf(fp, "Hello World!\n");
    fclose(fp);

    fp = fopen(FNAME, "r");
    if(fp == NULL){
        fprintf(stderr, "Cannot read file %s ?\n", FNAME);
        exit(EXIT_FAILURE);
    }
    char str[BIGSTR];
    if(fgets(str, BIGSTR, fp) == NULL){
        fprintf(stderr, "Cannot read 1st line of %s ?\n", FNAME);
        exit(EXIT_FAILURE);
    }
    fclose(fp);
    // Newline is read too
    if(strcmp(str, "Hello World!\n")){
        fprintf(stderr, "1st line of %s not correct?\n", FNAME);
        exit(EXIT_FAILURE);
    }
    return EXIT_SUCCESS;
}
```

- If you write a file, it overwrites it from the beginning.
- You must fclose() your file pointers otherwise there is a memory leak.
- The statement exit() allows you to exit the code **anywhere**, not just in main.
- There are three files already open for you: stdin, stdout and stderr.
- Therefore printf(...) is just a shorthand for fprintf(stdout, ...)
- fscanf() could be used instead of fgets().

stderr

- To write to screen you'd generally use `stdout`, so why is there `stderr` ?
- It's fairly common, when running a program, to want to redirect the output to a file.
- For instance, if you were to type:

  `$ ls > myfiles.txt`

  this will list all your files into myfiles.txt
- The ">" at the prompt redirects the output (or "<" input) to file rather than screen.
- If something went wrong though, the user would never see the message.
- Therefore, `stderr` exists so that there is a stream to display warnings/errors to the user.

Interlude : argc/v

- A traditional C program has `int main(int argc, char* argv[]);`
- argc is the number of words typed on the command line to execute the program.
- argv is an array of **pointers to** chars - i.e. an array of strings.
- This is not a traditional $2D$ array of characters - it's a $1D$ array of pointers - each string could be a different length.
- This is sometimes known as a ragged-right or jagged array.

```
1   #include <stdio.h>
2
3   int main(int argc, char* argv[])
4   {
5
6       printf("You typed %i arguments\n", argc);
7       printf("The name of your executable is : %s\n", argv[0]);
8       for(int i=1; i<argc; i++){
9           printf("Argument %d is : %s\n", i, argv[i]);
10      }
11
12      return 0;
13  }
```

Execution :

```
$ ./usingargs -c doof groob
You typed 4 arguments
The name of your executable is : ./usingargs
Argument 1 is : -c
Argument 2 is : doof
Argument 3 is : groob
```

Back to Files : One Character at a Time

```c
// Some of the functionality of cp
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char* argv[])
{
    if(argc != 3){
        fprintf(stderr, "Usage : %s <filein> <fileout>\n", argv[0]);
        exit(EXIT_FAILURE);
    }
    FILE* fpin = fopen(argv[1], "r");
    if(!fpin){
        fprintf(stderr, "Cannot read from %s\n", argv[1]);
        exit(EXIT_FAILURE);
    }
    FILE* fpout = fopen(argv[2], "w");
    if(!fpout){
        fprintf(stderr, "Cannot write to %s\n", argv[2]);
        exit(EXIT_FAILURE);
    }

    char c;
    while((c = fgetc(fpin)) != EOF){
        fputc(c, fpout);
    }
    fclose(fpin);
    fclose(fpout);
    return EXIT_SUCCESS;
}
```

- This is a very basic version of the Linux command cp :

  $ cp oldfile.txt newfile.txt

- Almost all Linux programs access arguments typed on the command line.
- fgetc() and fputc() are the file equivalents of getchar and putchar.

Bulk Copying

- Copying one character at a time is very slow for large files.
- `fread()` and `fwrite()` will I/O many characters at once.
- Here we save an entire array to a binary file - another program could read this in later.

```c
/* Compute some factorials and save them
   for another program to read back later. */
#include <stdio.h>
#include <stdlib.h>

#define FACTS 20

typedef unsigned long facttype;

int main(int argc, char* argv[])
{
   if(argc != 2){
      fprintf(stderr, "Usage : %s <fileout>\n", argv[0]);
      exit(EXIT_FAILURE);
   }
   FILE* fpout = fopen(argv[1], "wb");
   if(!fpout){
      fprintf(stderr, "Cannot write to %s\n", argv[1]);
      exit(EXIT_FAILURE);
   }
   facttype facts[FACTS] = {1};
   for(facttype i=1; i<FACTS; i++){
      facts[i] = facts[i-1]*i;
   }

   int n = fwrite(facts, sizeof(facttype), FACTS, fpout);
   if(n != FACTS){
      fprintf(stderr, "Cannot write to %s\n", argv[1]);
      exit(EXIT_FAILURE);
   }
   fclose(fpout);
   return EXIT_SUCCESS;
}
```

Window/DOS vs. Unix Text Files

- Text files created on DOS/Windows machines have different line endings than files created on Unix/Linux.
- DOS uses *carriage return* and *line feed* (`"\r\n"`) as a line ending.
- Unix uses just *line feed* (`"\n"`).
- You should generally avoid assuming line endings - use a function that reads entire lines at once such as `fgets()`.
- When you open a file in textmode `fopen("file.txt", "rt")` some automic translation may be done on input/output.

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define BIGSTR 10000

int main(int argc, char* argv[])
{
   if(argc != 2){
      fprintf(stderr, "Usage : %s <file>\n", argv[0]);
      exit(EXIT_FAILURE);
   }
   FILE* fpin = fopen(argv[1], "rb");
   if(!fpin){
      fprintf(stderr, "Cannot read %s\n", argv[1]);
      exit(EXIT_FAILURE);
   }
   char str[BIGSTR];
   if(fgets(str, BIGSTR, fpin)==NULL){
      fprintf(stderr, "Cannot read %s\n", argv[1]);
      exit(EXIT_FAILURE);
   }
   int n = strlen(str);
   if(n > 1){
      if((str[n-1] == '\n') && (str[n-2] == '\r')){
         printf("Looks like a DOS file?\n");
      }
      else{
         printf("Looks like a Unix file?\n");
      }
   }
   fclose(fpin);
   return EXIT_SUCCESS;
}
```